SIH26261 — Idea Submission & Presentation Plan (v2)

Problem Statement: SIH26261 · *AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification* · AICTE, Ministry of Education

Theme / Category: Smart Education · Software

> **One-line pitch:** "The only SIH26261 solution that *proves* it works — measured forgery-detection accuracy on a labelled benchmark, real APH 2024-27 rules with clause citations, and dossier-level cross-checks that no KYC tool does."

Why v2 is different

The evaluation found that the visible SIH26261 field (≈23 public repos) already ships forensics, APH rules, issuer mocks, photo-reuse checks and human-in-the-loop review. Those are **table stakes**. Judges screen a 6-slide PDF without us in the room, so the deck must lead with what others **cannot show**:

#	Change	Where it lands in the deck
1	**Benchmark is the headline** — measured precision/recall on a labelled synthetic forgery set	Slides 2, 4, 5
2	**Real APH 2024-27 rules** with clause numbers (no placeholders)	Slides 2, 3, 6
3	**Dossier cross-consistency graph** — land deed ↔ building plan ↔ OC ↔ fire certificate ↔ declared area ↔ faculty	Slide 2 (hero visual)
4	**Security condensed into one "Trust Layer"** mapped to DPDP + MeitY AI Guidelines	Slide 3 (one box)
5	**Defined 36-hour MVP** — what we build vs. what is roadmap	Slide 4

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Part A — The 6-slide idea submission 1. Slide 1 — Title 2. Slide 2 — Proposed Solution 3. Slide 3 — Technical Approach 4. Slide 4 — Feasibility & Viability 5. Slide 5 — Impact & Benefits 6. Slide 6 — Research & References

Part B — Supporting detail (for the team, Q&A and the finale)

- 7. The Benchmark (headline)
- 8. APH 2024-27 Rule Pack
- 9. Dossier Cross-Consistency Graph
- 10. Trust Layer

- 11. 36-Hour MVP Scope
- 12. Process Flow
- 13. Differentiation vs. the Field
- 14. Deck Design Rules & Checklist
- 15. Work Split

Part A — The 6-Slide Idea Submission

Official template: **max 6 slides including title, PDF, points and diagrams — no paragraphs.** Every slide below lists the *exact content*, the *one visual*, and what *not* to put on it.

Slide 1 — Title

Field	Content
Problem Statement ID	**SIH26261**
PS Title	AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
Theme	Smart Education
Category	Software
Team ID / Name	*[fill in]* — avoid "Pramaan"/"Nirikshan"-style names already used by other teams
Solution name + tagline	*[Name]* — "Evidence, not opinions: measured AI pre-scrutiny for AICTE approvals"

Slide 2 — Proposed Solution

Headline (top of slide, large): *Pre-scrutiny that proves itself — measured on a labelled forgery benchmark.*

Hero visual: the **dossier cross-consistency graph** (Section 9) — one institution's documents as nodes, with a red edge where the fire certificate address ≠ the building-plan address.

Bullets (max 6):

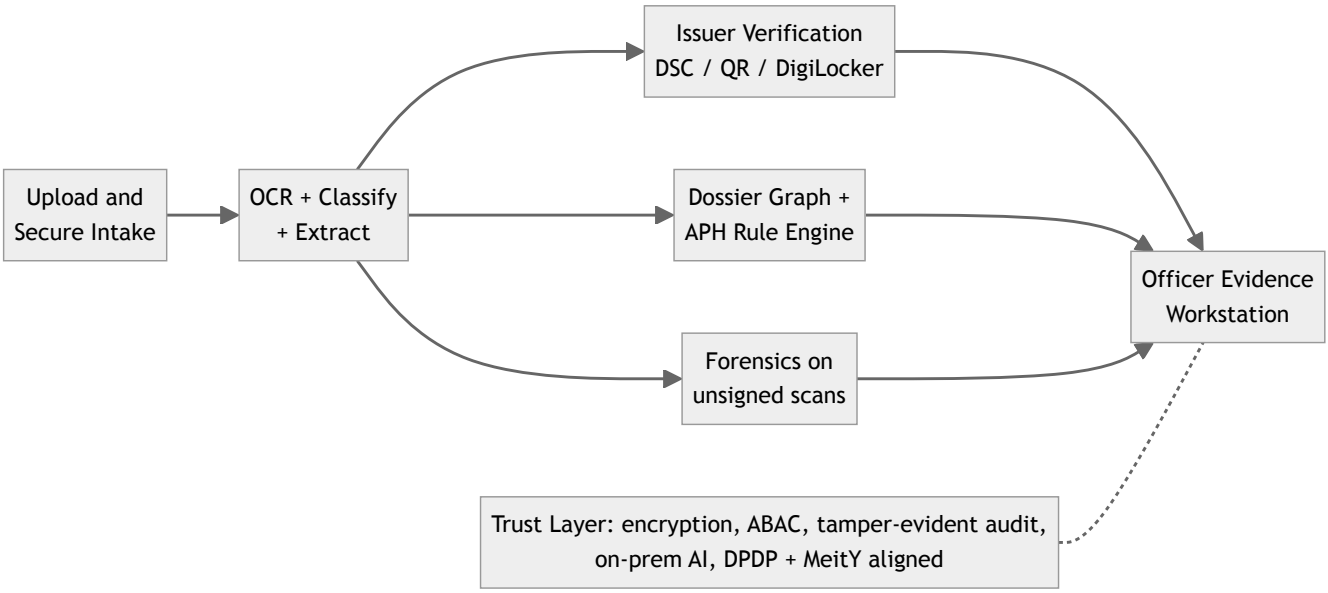
- **Dossier-level cross-checks:** land deed ↔ building plan ↔ Occupancy Certificate ↔ Fire Safety Certificate ↔ declared built-up area ↔ faculty roster — catches inconsistencies that single-document KYC tools cannot.
- **Real APH 2024-27 rules as code:** 20 rules with clause citations, e.g. *Cl. 6.17(a)* land lease ≥ 30 years, *Annexure-5* UG Engg faculty:student 1:20, *Cl. 6.17(h)* structural stability certificate valid 5 years.
- **Trust issuer data first, forensics second:** digitally signed / DigiLocker / QR documents are verified at source; tamper forensics run only on scans.
- **Measured, not claimed:** precision/recall per tamper type on our open synthetic benchmark (Section 7).

- **Officer stays in charge:** every flag = evidence crop + rule clause + suggested action; no automated adverse decision (APH cl. 2.4 "Nil Deficiency" compliance loop preserved).
- **Institution pre-check:** same engine runs before submission → fewer deficiency cycles.

Do not put here: the tech stack, security details, a long feature list.

Slide 3 — Technical Approach

Visual: one architecture diagram (simplified Section 12 flow) with **five pipeline boxes + one Trust Layer box**:



Compact stack line (one row of logos): Next.js · FastAPI · PostgreSQL · MinIO · Redis workers · PaddleOCR (EN + Hindi) · pyHanko · OpenCV · Keycloak + OPA · Vault · Docker/K8s

Bullets:

- Rules stored as **versioned YAML** with APH clause IDs → editable when APH changes or the pending VBSA Bill replaces AICTE norms.
- **Verification-first pipeline:** signed → verify at source; unsigned scan → PDF-structure + image forensics; result one of 6 statuses (Verified / Mismatch / Expired / Pending / Source Unavailable / Manual Review).
- All AI runs **on-prem**; no applicant data leaves Indian infrastructure.

Slide 4 — Feasibility & Viability

Visual (left): a **36-hour MVP vs Roadmap** two-column box (Section 11). **Visual (right):** a small **benchmark results card** — target metrics now, filled with measured numbers once built.

Bullets:

- **Built from proven open-source** components; no custom deep-learning training required.

- **Benchmark:** 200 synthetic institutions, ~2,000 documents, 8 labelled tamper types → precision, recall, false flags per genuine application.
- **Integration honesty:** DigiLocker requester access needs NeGD approval (OAuth + PKCE, India-hosted); we ship a **spec-faithful mock** and real offline DSC/QR verification.
- **Risks & mitigations:** forensics noisy on rescans → used only for unsigned docs and always advisory; no public fire/land APIs → adapters + manual-review status; rule changes → versioned rule packs with dry-run.

Slide 5 — Impact & Benefits

Visual: before/after pipeline strip — "*Self-disclosure trusted → deficiencies found after approval*" vs. "*Inconsistencies caught at submission, evidence attached.*"

Bullets:

- **For AICTE scrutiny officers:** triaged queue, evidence-linked flags, rule clause on every finding → less repetitive checking, more consistent decisions.
- **For institutions:** pre-check before submission → fewer deficiency rounds (EoA letters today list "deficiencies based on Self Disclosure" to be fixed within 6 months).
- **For EVC (inspection):** auto-generated "verify on site" list — only items the system could not confirm.
- **Integrity:** cross-institution reuse of photos/faculty detected using privacy-preserving hashes.
- **Accountability:** every rule run and officer action in a tamper-evident log → reproducible decisions for appeals and RTI.
- **Metrics we will report:** precision/recall per tamper type, false flags per genuine application, time per application (automated vs manual baseline).

Slide 6 — Research & References

- **AICTE Approval Process Handbook 2024-25 to 2026-27** — Cl. 2.3.5(b), 2.4, 6.14, 6.17(a)–(j), 6.18, 6.19, 7.4; Annexures 3, 4, 5, 14.
- **DPDP Act 2023 & DPDP Rules 2025** (notified Nov 2025) — purpose limitation, retention, security safeguards.
- **MeitY India AI Governance Guidelines (Nov 2025)** — human oversight, accountability, "understandable by design".
- **DigiLocker / API Setu** requester integration documentation (NeGD).
- **PDF-structure forensics:** arXiv 2507.00827; copy-move and noise-inconsistency forgery detection literature.
- **Prior SIH work:** SIH1729 *AI-supported AICTE approval portal*, SIH1730 *AI-driven inspection* (SIH 2024) — we extend these from workflow to verified, measured validation.

Part B — Supporting Detail

7. The Benchmark — Our Headline

No public SIH26261 project reports measured accuracy. We will.

Item	Design
Dataset	200 synthetic institutions → ~2,000 documents (land deed, building plan, OC/CC, fire safety cert, structural stability cert, affidavits, faculty roster, infrastructure photos)
Generator	Templated documents (Jinja → PDF), randomised fonts/stamps/signatures, Faker for Indian names/addresses, scanner-noise simulation (blur, JPEG, skew)
8 tamper types	1. altered date · 2. altered survey number / address · 3. swapped stamp/seal · 4. pasted signature · 5. edited PDF text layer (incremental save) · 6. photo reused across institutions · 7. faculty claimed by two institutions · 8. cross-document inconsistency (area / address mismatch)
Split	70% genuine, 30% tampered; held-out test set never seen during tuning
Metrics	Precision, recall, F1 **per tamper type**; **false flags per genuine application**; mean processing time per application
Release	Generator + dataset published on GitHub so judges can reproduce numbers

Deck rule: until measured, show *targets* clearly labelled as targets. Never present a number we have not measured.

8. APH 2024-27 Rule Pack (v1)

All clauses below were taken from the official **Approval Process Handbook 2024-25 to 2026-27** text.

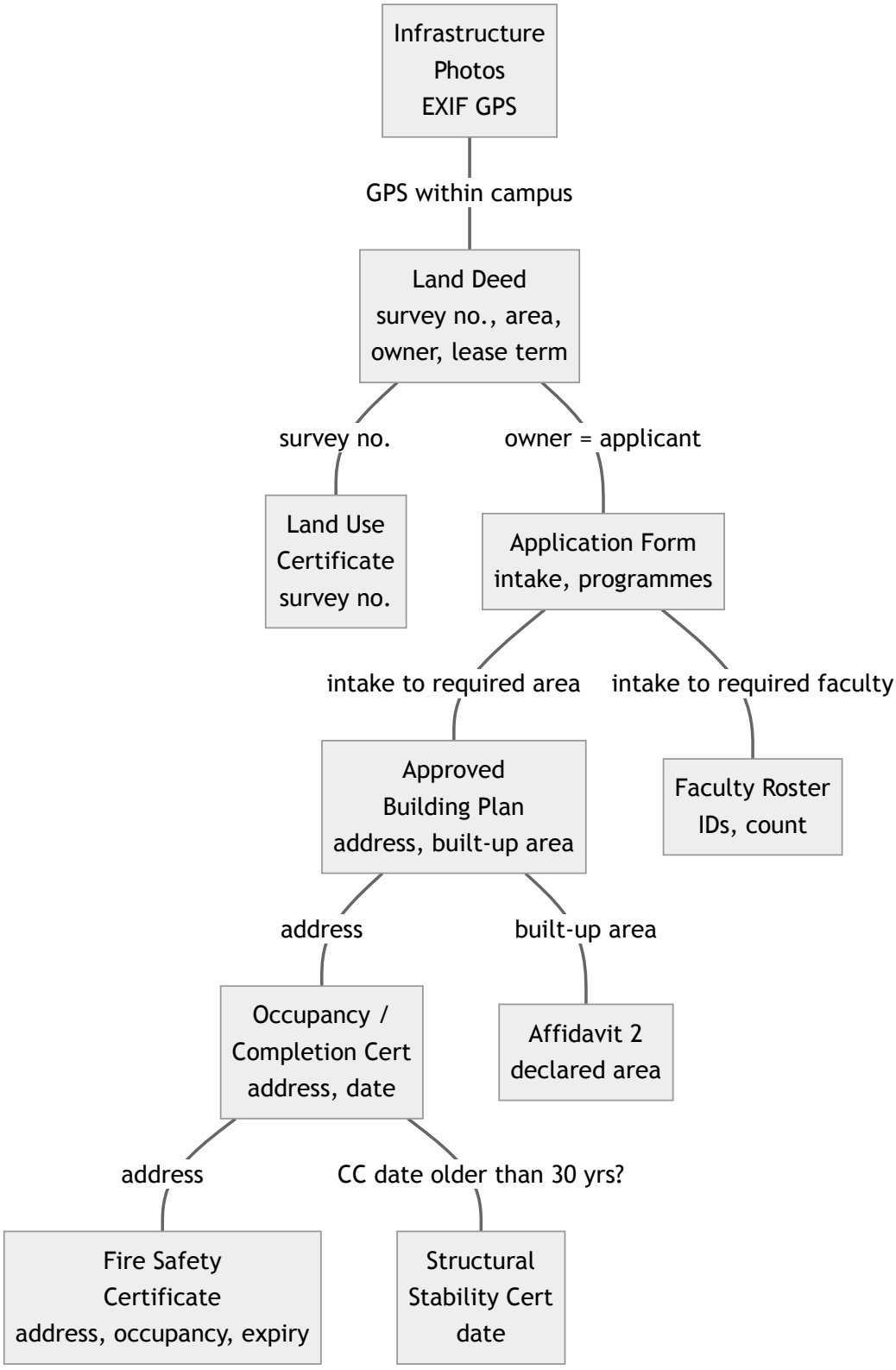
Rule ID	Check	APH reference
LAND_TITLE	Land in applicant's name, or registered lease ≥ 30 years	Cl. 6.17(a)
LAND_LIVE_LEASE	Remaining lease ≥ maximum programme duration at application	Cl. 6.17(a)
LAND_DEED_TYPE	Deed is a registered Sale / Settlement / Irrevocable Gift / Lease deed	Cl. 6.17(b)
LAND_LEASE_AFFIDAVIT	Joint Affidavit 7: lease irrevocable for 30 years (where leased)	Joint Affidavit 7 (APH document list)
LAND_CONTIGUOUS	Plots contiguous; obstacles need a Safety Certificate	Cl. 6.17(d)
LAND_USE_CERT	Land Use / Conversion certificate from competent authority present	Cl. 6.17(e)
BLDG_PLAN_MATCH	Building constructed as per approved plan; plan revalidated after major changes	Cl. 6.17(f)
BLDG_OC	Occupancy / Completion Certificate / Building Licence / Form D present	Cl. 6.17(g); penal ground Cl. 7.4(j)
BLDG_STRUCT_STAB	If > 30 years since Completion Certificate → Structural Stability Certificate present and ≤ 5 years old	Cl. 6.17(h)
BLDG_FIRE	Valid (unexpired) Fire Safety Certificate	Cl. 6.17(i)

Rule ID	Check	APH reference
AUTHORITY_VALID	Certificate issuer is the state's designated competent authority	Cl. 6.17(j), Annexure-14
BUA_SUBCATEGORY	Instructional, administrative, amenities, circulation areas each meet norms	Cl. 6.18, Annexure-3
BUA_CLASSROOMS	1 classroom per division per year, 66 m² (Diploma/UG)	Annexure-3
BUA_AFFIDAVIT2	Affidavit 2 declares built-up area unchanged; compare with plan/OC area	Affidavit 2; Cl. 6.39
FAC_RATIO_UG	UG Engineering & Technology faculty:student ≥ 1:20 (on approved intake)	Cl. 6.14, Annexure-5 (5.2)
FAC_RATIO_DIP	Diploma Engineering faculty:student ≥ 1:25	Cl. 6.14, Annexure-5 (5.1)
FAC_ID_UPDATE	Faculty IDs added/removed on portal within the month of joining/leaving	Cl. 6.14(c)
FAC_SALARY_ECS	Faculty salary paid via ECS; salary statements match roster	Cl. 6.14(a)
LIB_COMP	Computers, software, books, library norms met	Cl. 6.19, Annexure-4
FRAUD_ESCALATE	Any document confirmed fraudulent → escalate to officer (criminal-case clause)	Cl. 2.3.5(b)

Each rule is stored as YAML: `rule_id`, `version`, `applies_to`, `check`, `severity`, `aph_ref`, `on_fail` message (English + Hindi).

9. Dossier Cross-Consistency Graph

The core novelty. Each document becomes a node of extracted facts; edges are rules that must agree.



What it catches (examples for the demo):

- Fire certificate issued for a different address than the building plan.
- Affidavit 2 declares 12,000 m² but the approved plan shows 9,800 m².
- Building > 30 years old with no structural stability certificate.
- Faculty roster below 1:20 for the approved intake.
- Infrastructure photo GPS outside the land parcel, or the same photo used by another institution.

10. Trust Layer (Security & Privacy, condensed)

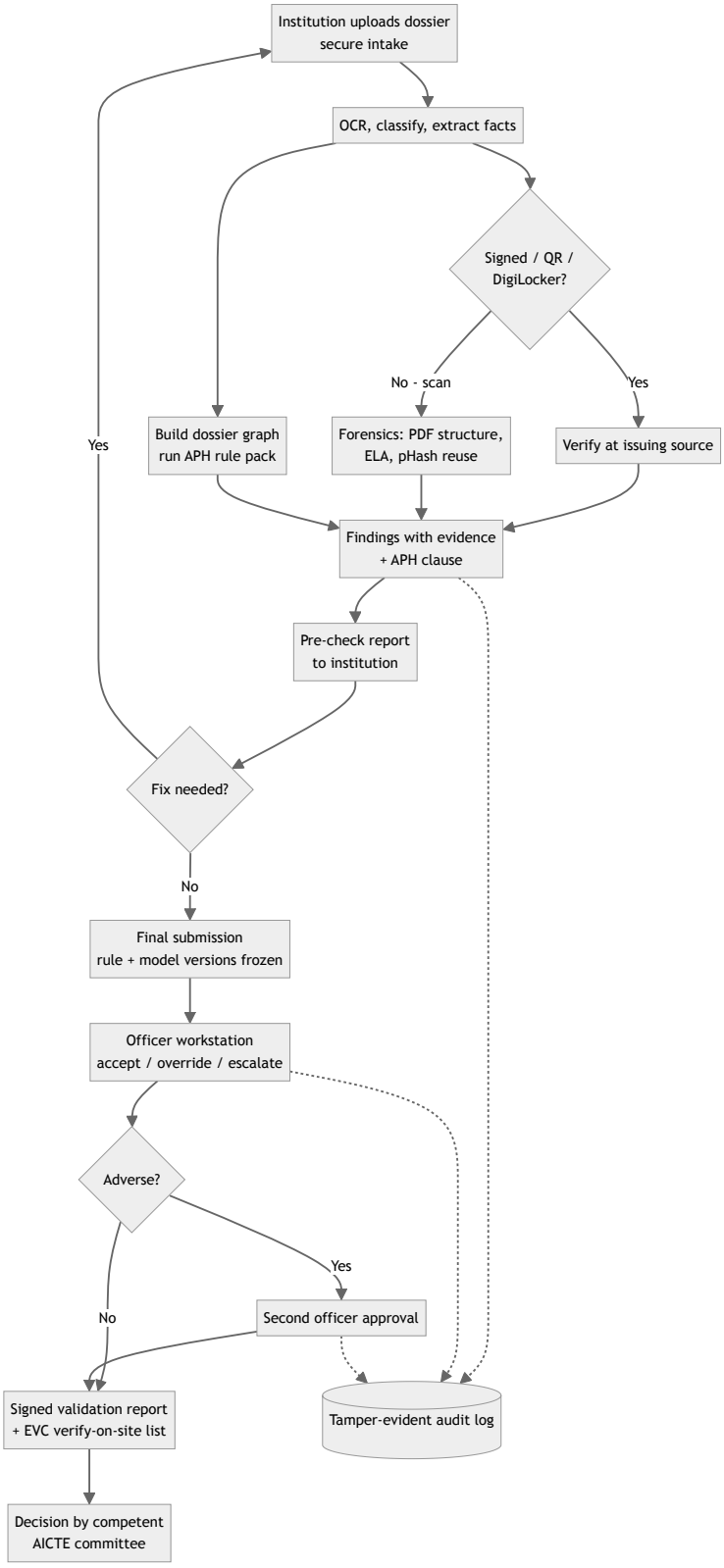
On the deck this is **one box**. In the build it is real:

Control	Implementation	Maps to
Encryption	Per-document AES-256-GCM keys wrapped by Vault; PII fields tokenised	DPDP security safeguards
Access	Keycloak MFA + OPA attribute rules (region, role, scrutiny window)	DPDP; least privilege
Tamper-evident audit	Hash-chained log + signed periodic root; one-click integrity check	MeitY: accountability
Human oversight	Four-eyes approval for any adverse outcome	MeitY: people first / human oversight
Privacy-preserving matching	Faculty IDs / PAN compared as keyed HMAC hashes	DPDP: data minimisation
Safe uploads	ClamAV + PDF disarm & rebuild, magic-byte checks	Secure by design
Sovereign AI	All models on-prem; no third-party AI APIs	Data localisation

11. 36-Hour MVP Scope

Build in 36 hours (demo)	Roadmap (slide mention only)
Upload + classify 8 document types	All APH application categories
OCR + extraction (English; Hindi for 2 doc types)	Full Indic coverage, Bhashini translation
Dossier graph + 20 APH rules	No-code rule builder with dry-run
DSC (pyHanko) + QR verification, DigiLocker mock	Live DigiLocker / NAD / state portal integration
Forensics: PDF structure + ELA + pHash on unsigned scans	Noiseprint-class deep forensics
Benchmark generator + measured metrics	Larger benchmark, external validation
Officer workstation: evidence viewer, 6 statuses, accept/override	Triage score, inspection planner, remote EVC app
Trust Layer: Keycloak, encrypted storage, hash-chained audit	Full OPA policy set, Vault HA, Kubernetes

12. Process Flow



13. Differentiation vs. the SIH26261 Field

Capability	Typical SIH26261 repo	Ours
OCR + forensics	✅ common	✅ (but only on unsigned scans)

Capability	Typical SIH26261 repo	Ours
APH rules	Some, often few or placeholder clauses	**20 rules, real clause citations**
Single-document checks	✅ common	✅
Dossier cross-consistency graph	Rare / partial	✅ **core feature**
Measured accuracy	❌ none found	✅ **benchmark, per tamper type**
Issuer verification	Mocks	Real DSC/QR + spec-faithful DigiLocker mock
Security	Basic auth	**Trust Layer mapped to DPDP + MeitY**
Demo honesty	Often over-claims integrations	Explicit "mock vs real" labels

14. Deck Design Rules & Submission Checklist

Design rules

- One message per slide, stated as the slide headline.
- Max ~6 bullets per slide, ≤ 12 words each. Diagrams over text.
- Every claim either cites an APH clause or a measured number.
- Use the words **"indicator"** and **"decision support"** — never "detects forgery" or "approves applications".
- No blockchain, no "LLM decides".
- Consistent colour coding: green = verified, amber = manual review, red = mismatch.
- Export to PDF; check fonts embedded and diagrams readable at 100% zoom.

Checklist before upload

- [ ] Official SIH template used; ≤ 6 slides including title
- [ ] PS ID SIH26261, theme, category, team ID correct on Slide 1
- [ ] Slide 2 hero = dossier graph; benchmark mentioned in headline
- [ ] Slide 3 = one diagram + one-line stack + Trust Layer box
- [ ] Slide 4 = MVP vs roadmap + benchmark card + risks
- [ ] Slide 5 = stakeholder impact + metrics we will report
- [ ] Slide 6 = APH clauses, DPDP, MeitY, DigiLocker, forensics papers
- [ ] No unmeasured numbers presented as results
- [ ] Team composition rules met (6 members, ≥ 1 woman)
- [ ] (If allowed) short demo video link showing the dossier-graph catch

15. Work Split (presentation-first)

Track	Owner focus	Deck output
1. Deck lead	Storyline, visuals, final PDF	All slides
2. Benchmark	Generator, tamper types, metrics script	Slide 4 card, Slide 5 metrics

Track	Owner focus	Deck output
3. Rules + Dossier graph	YAML rule pack, graph checks	Slide 2 hero, Slide 6 clauses
4. Verification + Forensics	pyHanko, QR, DigiLocker mock, PDF/ELA/pHash	Slide 3 pipeline
5. Trust Layer	Keycloak, encryption, audit chain	Slide 3 box, Q&A backup
6. Workstation UI + Demo	Evidence viewer, demo video	Screenshots for Slides 2–5

All AI outputs are decision-support only. Statutory scrutiny, inspections and final regulatory decisions remain with the competent AICTE authorities.